

Smart Study Generator Agent - Knowledge Base

Introduction

Smart Study Generator is an AI-powered study assistant that helps students learn efficiently using IBM watsonx.ai, IBM Granite Models, LangFlow, IBM Cloud and Retrieval-Augmented Generation (RAG).

Features

Summarizes notes, generates flashcards, creates quizzes, explains concepts, answers academic questions, creates personalized study plans and tracks learning progress.

Supported Resources

PDF notes, textbooks, lecture notes, research papers, presentations and study guides.

Technologies

IBM watsonx.ai for AI, IBM Granite Models for reasoning, LangFlow for workflow orchestration, RAG for accurate retrieval, Vector Database (FAISS/Chroma) for semantic search.

How It Works

Users upload study material. The system indexes the content, retrieves relevant information using RAG, and generates personalized responses, summaries, quizzes and flashcards.

Benefits

Saves time, improves understanding, supports exam preparation, offers personalized learning and is available 24/7.

Limitations

Response quality depends on uploaded materials and should complement, not replace, teachers.

Future Scope

Voice interaction, multilingual support, AI mock tests, LMS integration and advanced analytics.